Exercise series 1: Solutions

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University of Antwerp

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Problem 1

CREATE AN E.R. MODEL FOR THE FOLLOWING SCENARIO

A travel organization is responsible for the organization of holidays and selling them to customers. To support their business, they need a database to maintain all the different travel packages, and customer bookings. A travel package consists of flight and residence. For the flight to and from, we store the departure and destination location, date and time, the name of the flight carrier (e.g. "Ryanair") and the type of seat ("economy", "plus" or "business"). For a flight we also maintain the price, with a different rate for adults and children. For the residence (e.g. hotel or bungalow park) we maintain its unique code, name, description, address, and the total number of rooms available. Each residence has one or more rooms and each room is uniquely identified using the combination of residence code and room. For a room we also maintain its description, number of beds available, and the price per night. The booking consists of a selected travel package offer. A booking is associated with one or more persons, where one person is the main contact for the booking. A person is identified by his/her name, date of birth and adult status (true or false). A booking is also associated with a selection of one or more rooms at the residence associated with the selected travel package. Remark that the same travel package can be booked by multiple groups of persons, for example group "G1" with 4 persons books "TP1" with room "R001" and "R002" at Barcelona hotel "Costa-Barca", and group "G2" with 2 persons, also books "TP1", but at room "R003".

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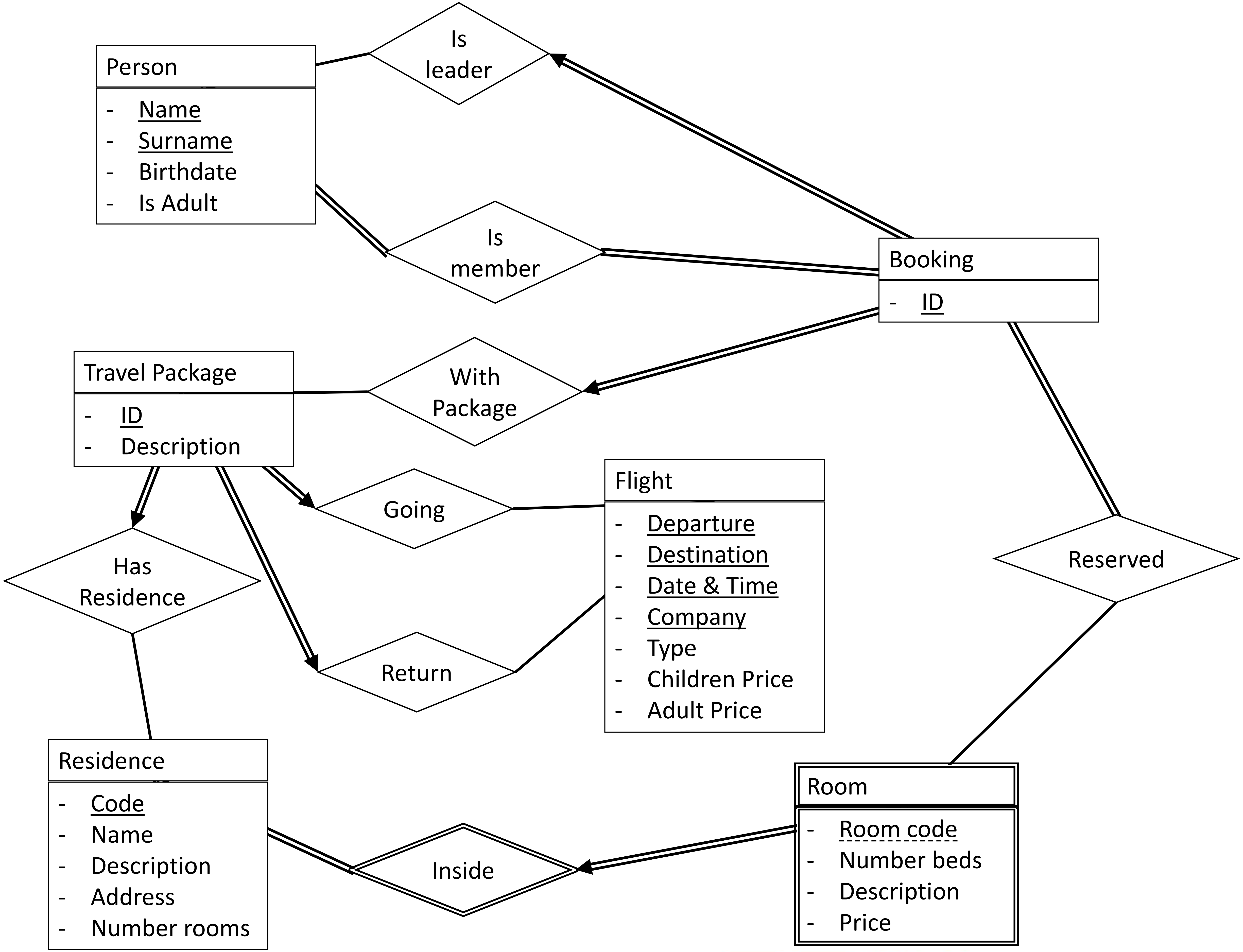

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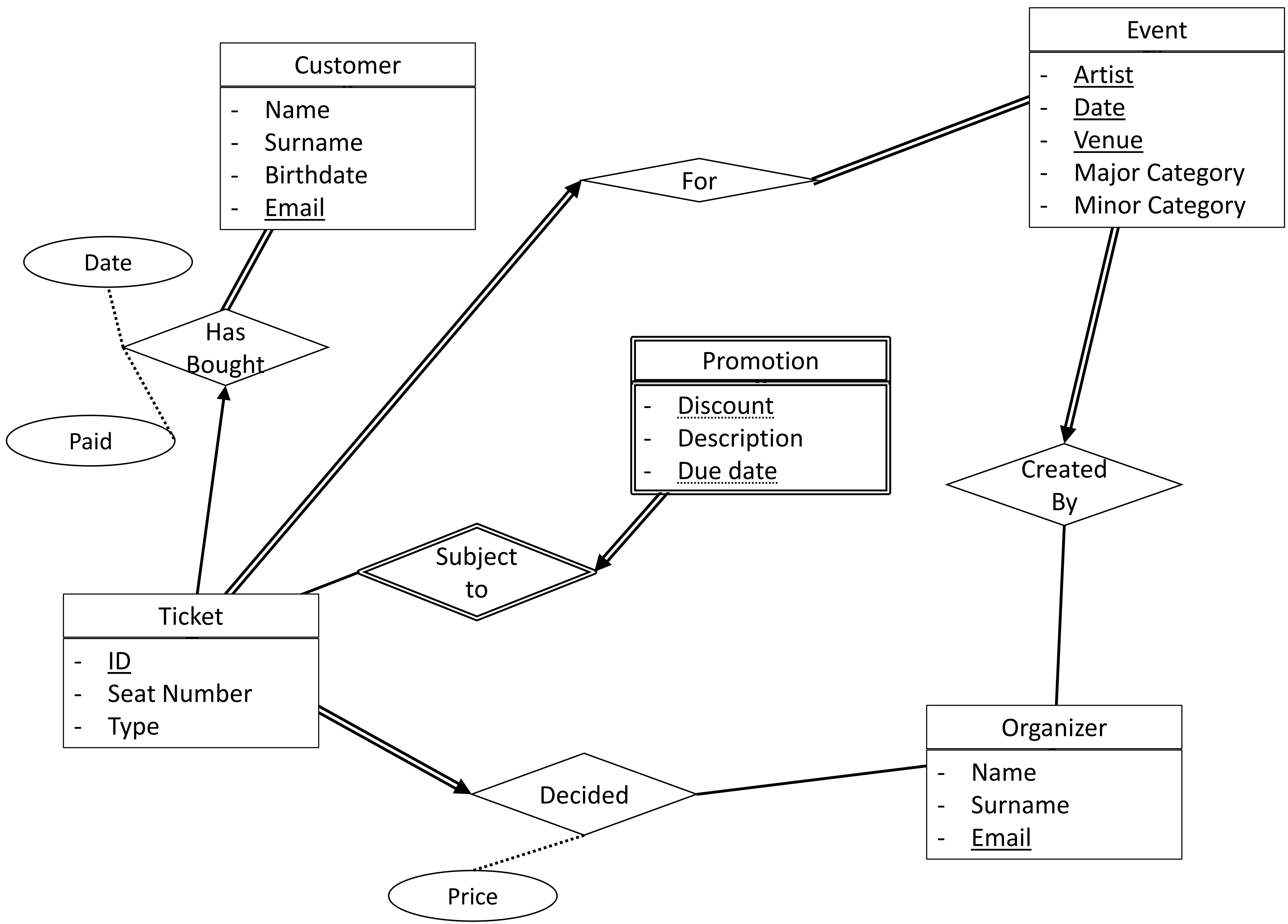
Problem 2

CREATE AN E.R. MODEL FOR THE FOLLOWING SCENARIO

A ticket booking platform is a marketplace where customers can buy tickets for events, such as concerts, shows and sport events. Different organizers can create events. An organizer logs on to the system, with his or her account, and can create one or more events. An event has a major (e.g. "concert") and minor category (e.g. "pop-rock") and associated artist name (e.g. "Angèle"). An event is uniquely identified by the combination of the artist name, date, and venue. It is possible that an event is repeated at the same venue (e.g. "Ancienne Belgique" in Brussel) on different dates, or repeated at different venues on different dates, for example, when an artist goes on a tour. For each event there is a number of available tickets. Different types of tickets are possible, e.g. VIP seats are more expensive than standing seats. Remark that, for each type of ticket the price is set by the organizer of the event. For some events there's an early registration promotion, that is, if tickets are purchased before a certain date there's a discount of x%. Each ticket has a unique code, a seat number and a ticket type. A customer has a first and last name, birth date and unique e-mail address. Each customer can buy one or more available tickets for an event. When a ticket is bought, the date of purchase and price is registered by the system. Each customer can buy at most 10 tickets for a single event.



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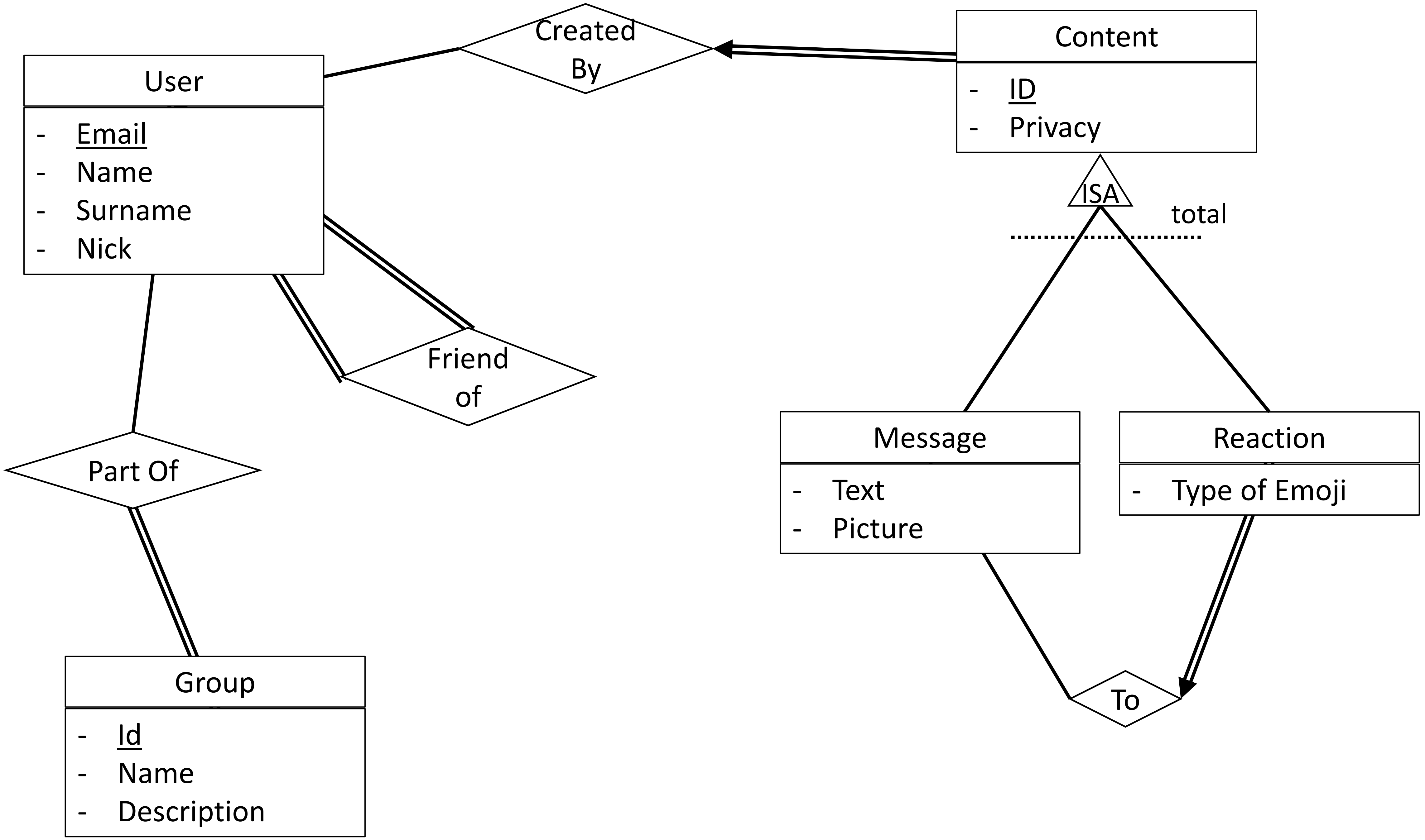
Problem 3

CREATE AN E.R. MODEL FOR THE FOLLOWING SCENARIO

A social network (e.g. Instagram) consists of a network of users. Your task is to model the basics of a social network and propose interesting extensions. Each user has one or more friends. A user can also belong to zero or more groups. A user can post a message containing a picture and a text message. Each message is either public, for the entire social network community, or only available for friends. Other users (or friends) can react to posts with a message and/or like the post. Users can also react to a reaction.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